

ANDRE NUGRAHA AKBAR

Supply Chain Data Analyst | Demand Planning & Inventory Optimization

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SUMMARY

Industrial Engineering graduate specializing in supply chain analytics, combining inventory planning domain knowledge with applied machine learning and operations research (ILP, demand forecasting, decision support systems). Experienced in translating business requirements into data-driven decision support tools using Python and Power BI. Seeking a full-time Supply Chain Data Analyst / Demand Planning role to drive operational efficiency through predictive analytics.

EXPERIENCE

Freelance Data Visualization Analyst

Sep 2025 – March 2026

Project-Based

- Developed interactive Power BI dashboards to translate client business needs into operational analysis and strategic decision-making tools.
- Processed, cleaned, and transformed raw operational data into tactical visual insights that were easily understood by non-technical stakeholders.
- Collaborated with clients to ensure the resulting analytical reports aligned with their business key performance indicators (KPIs).

Information Systems & Decision Laboratory (LSIK FT-UNAND)

Jun 2024 – Jun 2025

Lab Assistant & HRD Staff • Padang, West Sumatra

- Led practicum sessions across 4 courses per year (Engineering Drawing, Information Systems, Computer Programming, Data Analytics), mentoring 24-48 students annually in data-driven case studies.
- Acted as project mentor, guiding students from business problem identification to technical model design.
- Managed recruitment and onboarding for laboratory assistants as part of HRD function.

PT Semen Padang

Jan – Feb 2024

Inventory Division Intern • Padang, West Sumatra

- Validated and reconciled 1,000+ spare part records against SAP system over a 1-month internship, identifying and correcting inventory location mismatches.
- Updated material location and identification data in the warehouse system, improving traceability for daily stock audits covering 10–50 items per day.

ORGANIZATION EXPERIENCE

DOTCOM HMTI FT UNAND

Aug 2023 – Aug 2024

HRD Department Coordinator • Padang, West Sumatra

- Managed the end-to-end recruitment process for new division members, successfully screening, selecting, and onboarding all 9 applicants (100% acceptance rate) into the organization.
- Designed training curriculum focused on data science and analytical thinking for organization members.

Himpunan Mahasiswa Teknik Industri (HMTI FT UNAND)

Research & Development Department Staff • Padang, West Sumatra

Mar 2023 – Dec 2023

- Supported research activities and competency development programs for students.
- Contributed to knowledge-sharing initiatives and technical skill enhancement.
- Assisted internal collaboration to improve departmental program effectiveness.

PROJECTS

Multi-Depot Distribution Optimizer DSS

Python, ILP, CVRP, Streamlit • [Live Demo](#)

- Built a 2-stage optimization DSS (Integer Linear Programming + Capacitated Vehicle Routing Problem) for order allocation and route planning across 4 distribution centers.
- Reduced total assignment distance by 57.5% (38,270 km → 16,273 km) and cut utilization imbalance by 12.2% in a 7-day simulated test (19,420 orders, 650 customers).
- Eliminated all vehicle capacity overloads post-optimization, with average solve time of 0.379s/day enabling practical daily use.

Smart Inventory Management System

Python, XGBoost, Pandas, Streamlit • [Live Demo](#)

- Developed XGBoost forecasting model for 1,163 SKUs (296K transaction records), using lag/rolling/temporal feature engineering to achieve 5–10% MAPE, outperforming Naive and Moving Average baselines.
- Integrated forecasts into EOQ, Reorder Point, and Safety Stock calculations to reduce stockout and overstock risk.
- Built interactive Streamlit dashboard for inventory monitoring and decision support.

Supplier Performance Analytics Dashboard

Python, XGBoost, SVR, Streamlit • [Live Demo](#)

- Designed a dual-model supplier risk system (SVR for lead time regression, XGBoost for PO delay classification) with a 3-tier risk recommendation engine.
- Built dual-validation logic cross-checking model predictions against rule-based thresholds to flag low-confidence cases for manual review.
- Engineered supplier-level historical features (reliability, late rate, lead time variance) with Target Encoding for high-cardinality categorical fields.

EDUCATION

Universitas Andalas — Bachelor of Industrial Engineering

Aug 2021 – Jul 2025

GPA: 3.62 / 4.00

- Thesis: "Development of a Demand Forecasting System for Reorder Point Calculation Using Machine Learning in the Retail Sector (Case Study: XYZ Mart)"

SKILLS

Data Analysis & Visualization: Python, Pandas, SQL, Power BI

Supply Chain & Inventory: Demand Forecasting, EOQ, Reorder Point, Safety Stock

Operations Research: Integer Linear Programming (ILP), CVRP, Route Optimization

Systems & Tools: SAP (Inventory), Microsoft Office, SPSS